

Azure Honeypot + Live Threat Map

Security Lab Report

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Total Attacks	Countries Detected	Monitoring Period	Peak (single IP)
1,600+	8	24 hours	799 attempts

1. Executive Summary

This report documents the deployment and operation of a cloud-based honeypot on Microsoft Azure as part of a cybersecurity portfolio project. The honeypot — a deliberately vulnerable Ubuntu Server virtual machine with all inbound ports open to the public internet — was monitored using Microsoft Sentinel SIEM over a 24-hour period.

Within minutes of deployment, automated threat actors began probing the system. Over the monitoring period, more than 1,600 failed SSH login attempts were recorded from over 20 unique IP addresses across 8 countries. All attack data was ingested into Microsoft Sentinel, analysed using KQL queries, and visualised on a live geolocation threat map.

This project demonstrates practical skills in cloud infrastructure deployment, SIEM configuration, log analysis, KQL query writing, and threat visualisation — core competencies for a SOC Analyst or Cloud Security Engineer role.

2. Project Overview

Field	Details
Objective	Deploy a honeypot VM to attract and analyse real-world cyberattacks
Platform	Microsoft Azure (Free Trial — \$200 credit)
VM Image	Ubuntu Server 22.04 LTS
VM Size	Standard_D2als_v6
Region	East US
SIEM	Microsoft Sentinel (Workspace: law-honeypot)
Monitoring Agent	Azure Monitor Agent + Syslog Data Collection Rule
Duration	24 hours active collection

3. Methodology

Step 1 — Resource Group

Created resource group rg-honeypot-lab in East US to contain all project resources.

Step 2 — VM Deployment

Deployed Ubuntu Server 22.04 LTS VM with password authentication and SSH port open.

Step 3 — Internet Exposure

Added NSG inbound rule DANGER_ALL_INBOUND to allow all traffic on all ports from any source, making the VM visible to internet-wide scanners.

Step 4 — Log Analytics

Created Log Analytics Workspace law-honeypot as the central log repository for Sentinel.

Step 5 — Sentinel Setup

Enabled Microsoft Sentinel on the workspace, activating the 31-day free SIEM trial.

Step 6 — Data Collection

Created a Data Collection Rule connecting honeypot-vm to the workspace via Azure Monitor Agent, collecting Linux Syslog auth logs.

Step 7 — KQL Analysis

Wrote KQL queries to detect failed SSH logins, extract attacker IPs, summarise attack volumes by hour, and geolocate attackers.

Step 8 — Threat Map

Built a Microsoft Defender for Cloud Workbook with Map visualisation, plotting attacker geolocations sized by attack volume.

4. Key Findings

4.1 Attack Volume by Hour

Time (UTC)	Failed Logins	Notes
09:00	42	Early scanning activity
12:00	1	Low activity period
13:00	143	Increased brute force
14:00	751	PEAK — coordinated botnet activity
15:00	32	Sustained scanning
16:00	35	Continued activity
17:00	37	Evening scanning
18:00	20	Ongoing attempts

4.2 Top Attacking IP Addresses

IP Address	Failed Logins	Country	City
45.156.87.254	799	The Netherlands	Eygelshoven
45.156.87.13	799	The Netherlands	Eygelshoven
45.153.34.149	355	The Netherlands	Eygelshoven
178.175.167.68	150	Moldova	Anenii Noi
45.148.10.152	46	The Netherlands	Amsterdam
45.148.10.141	44	The Netherlands	Amsterdam
92.118.39.236	43	Romania	—
45.227.254.170	42	Panama	—
111.228.35.169	20	China	—

4.3 Countries of Origin

Country	Observation
The Netherlands	Majority of attacks — likely VPS/botnet hosting infrastructure

Moldova	High volume sustained brute force
Romania	Sustained scanning activity
China	Multiple IPs, moderate volume
Panama	Single IP, sustained attempts
Belgium	Low volume scanning
Germany	Moderate volume
France	Single attempt recorded

5. Analysis

5.1 Attack Pattern

All recorded attacks were SSH brute force attempts targeting the root account. This is consistent with automated botnet scanning — threat actors continuously scan the entire IPv4 address space for open port 22 and attempt common username and password combinations.

5.2 Botnet Indicators

- Multiple IPs within the same subnet (45.148.10.x, 45.156.87.x) indicate a single actor controlling multiple machines.
- Peak volume of 751 attempts in one hour indicates automated tooling such as Hydra or Medusa.
- Geographic spread across 8 countries suggests use of distributed VPS infrastructure or VPN exit nodes.
- No successful logins were recorded — the VM password was sufficiently strong.

5.3 Real-World Relevance

This experiment confirms that any internet-exposed SSH service will be discovered and attacked within minutes. Organisations must implement controls including: SSH key-based authentication, Just-in-Time VM access, IP allowlisting for administrative ports, and intrusion detection systems.

6. KQL Queries Used

Failed Login Detection

```
Syslog
| where Facility == "auth"
| where SyslogMessage contains "Failed password"
| parse SyslogMessage with * "from " AttackerIP " port" *
| where AttackerIP != ""
| summarize FailedLogins = count() by AttackerIP
| order by FailedLogins desc
```

Threat Map Geolocation Query

```
Syslog
| where Facility == "auth"
```

```

| where SyslogMessage contains "Failed password"
| parse SyslogMessage with * "from " AttackerIP " port" *
| where AttackerIP != ""
| summarize FailedLogins = count() by AttackerIP
| extend GeoInfo = geo_info_from_ip_address(AttackerIP)
| extend Latitude = toreal(GeoInfo.latitude)
| extend Longitude = toreal(GeoInfo.longitude)
| extend Country = tostring(GeoInfo.country)
| extend City = tostring(GeoInfo.city)
| project AttackerIP, FailedLogins, Latitude, Longitude, Country, City

```

7. Skills Demonstrated

Skill Area	What Was Done
Cloud Infrastructure	Deployed Azure VM, NSG, Resource Group from scratch
SIEM Configuration	Enabled and configured Microsoft Sentinel on Log Analytics
Log Ingestion	Set up Azure Monitor Agent and Data Collection Rules
KQL Query Writing	Wrote 6 custom queries for detection and threat analysis
Threat Visualisation	Built live geolocation threat map workbook
Threat Intelligence	Identified botnet patterns from log data
Security Monitoring	24-hour continuous monitoring and attack analysis
Documentation	Produced lab report, GitHub README, and KQL script files

8. Lessons Learned

- Cloud VMs exposed to the internet are attacked within minutes — hardening is not optional.
- SSH brute force is one of the most persistent and automated attack vectors against Linux.
- Microsoft Sentinel is a powerful and accessible SIEM for log analysis and visualisation.
- KQL is an intuitive query language that makes security log analysis fast and effective.
- Botnet activity is identifiable through subnet clustering patterns in authentication logs.
- Always set Azure budget alerts before deploying resources to avoid unexpected charges.
- Documenting a lab thoroughly is as valuable as building it for portfolio purposes.

9. Conclusion

This honeypot project successfully demonstrated the persistent and automated nature of cyber threats targeting internet-exposed infrastructure. Over 1,600 real-world attack attempts were captured, analysed and visualised within 24 hours using Microsoft Azure and Sentinel SIEM.

The project provided hands-on experience with cloud deployment, SIEM configuration, log analysis, and KQL query writing — skills directly applicable to SOC Analyst and Cloud Security Engineer roles. The threat map produced serves as a compelling visual demonstration of real-world attack

activity for portfolio and interview purposes.